WEEK 10

Illustration of Array

Associative Array

Multidimensional Array





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How to Create Array



WHAT IS ARRAY?

An array is a special variable, which can hold more than one value at a time.
If you have a list of items (a list of courses, for example), storing the course in single variables could look like this:

```
$courses1="BSIT";
$courses2="BSHRM";
$courses3="ACT";
```

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How to Create Function



WHAT IS ARRAY?

However, what if you want to loop through the list of courses and find a specific one? And what if you had not 3 course, but 10?

The solution is to create an array!
An array can hold many values under a single name, and you can access the values by referring to an index number.

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ARRAY



CREATING AN ARRAY

In PHP, the array() function is used to create an array:
array();

In PHP, there are three types of arrays:

- **Indexed arrays** - Arrays with numeric index
- **Associative arrays** - Arrays with named keys
- **Multidimensional arrays** - Arrays containing one or more arrays

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INDEX ARRAY

The index can be assigned automatically (index always starts at 0):

```
$courses=array(“BSIT",“BSHRM",“ACT");
```

or the index can be assigned manually:

```
$cars[0]="Volvo";
$cars[1]="BMW";
$cars[2]="Toyota";
```

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INDEX ARRAY

The following example creates an indexed array named $courses, assigns three elements to it, and then prints a text containing the array values:

```
<?php
$courses=array(“BSIT","BSHRM",“ACT");
echo “List of courses " . $courses[0] . ", " . $courses[1] . " and " .
$courses[2] . ".";
?>
```

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INDEX ARRAY - Looping

To loop through and print all the values of an indexed array, you could use a for loop, like this:

```
<?php
$courses=array(“BSIT",“BSHRM",“ACT");
$arrlength=count($courses);

for($x=0;$x<$arrlength;$x++)
  {
  echo $courses[$x];
  echo "<br>";
  }
?>
```



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ASSOCIATIVE ARRAY

Associative arrays are arrays that use named keys that you assign to them.
There are two ways to create an associative array:

$age=array(“John"=>“25",“Fred"=>“27",“Max"=>“63");

$age[‘John']=“25";
$age[‘Fred']=“27";
$age[‘Max']=“63";

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ASSOCIATIVE ARRAY



Example

```
<?php
$age=array(“John"=>“25",“Fred"=>“27",“Max"=>“63");
echo "Peter is " . $age['Peter'] . " years old.";
?>
```

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LOOP THROUGH ASSOCIATIVE ARRAY

To loop through and print all the values of an associative array, you could use a foreach loop, like this:

```
<?php
$age=array(“John"=>“25",“Fred"=>“27",“Max"=>“63");
foreach($age as $x=>$x_value)
  {
  echo "Key=" . $x . ", Value=" . $x_value;
  echo "<br>";
  }
?>
```

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MULTIDIMENSIONAL ARRAY

A multidimensional array is an array containing one or more arrays.
PHP supports multidimensional arrays that are two, three, four, five, or more levels deep. However, arrays more than three levels deep are hard to manage for most people.

The dimension of an array indicates the number of indices you need to select an element.
For a two-dimensional array you need two indices to select an element
For a three-dimensional array you need three indices to select an element

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Example:

```
<?php
echo $cars[0][0].": In stock: ".$cars[0][1].", sold: ".$cars[0][2].".<br>";
echo $cars[1][0].": In stock: ".$cars[1][1].", sold: ".$cars[1][2].".<br>";
echo $cars[2][0].": In stock: ".$cars[2][1].", sold: ".$cars[2][2].".<br>";
echo $cars[3][0].": In stock: ".$cars[3][1].", sold: ".$cars[3][2].".<br>";
?>
```